

# SemiFA: An Agentic Multi-Modal Framework for Autonomous Semiconductor Failure Analysis Report Generation

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**Abstract**—Semiconductor failure analysis (FA) is a labor-intensive process in which engineers examine inspection images, correlate equipment telemetry, consult historical defect records, and write structured reports. A single FA report typically requires 2–4 hours of domain expert time. We present SEMIFA, an agentic multi-modal framework that autonomously generates structured FA reports from semiconductor inspection images in under one minute. SEMIFA decomposes the FA process into a four-agent pipeline orchestrated by LangGraph: a *Defect-Describer* that classifies and narrates defect morphology using DINOv2 embeddings and LLaVA-1.6, a *RootCauseAnalyzer* that fuses equipment telemetry from SECS/GEM with historically similar defects retrieved from a Qdrant vector database, a *SeverityClassifier* that assigns severity and estimates yield impact, and a *RecipeAdvisor* that proposes corrective process parameter adjustments. A fifth node assembles the outputs into a PDF report. We introduce SEMIFA-930, a dataset of 930 annotated semiconductor defect images paired with structured FA narratives across nine defect classes, drawn from procedural synthesis, WM-811K, and MixedWM38. On SEMIFA-930, our DINOv2-based classifier achieves 92.1% overall accuracy on 140 validation images (macro F1 = 0.917), and the full pipeline produces complete FA reports in 48 seconds on an NVIDIA A100-SXM4-40 GB GPU. Qualitative evaluation demonstrates that Qdrant retrieval augmentation yields more specific root cause hypotheses referencing actual equipment history compared to retrieval-free baselines. To our knowledge, SEMIFA is the first system to integrate SECS/GEM equipment telemetry into a vision-language model pipeline for autonomous FA report generation.

**Index Terms**—semiconductor inspection, failure analysis, vision-language model, multi-agent system, DINOv2, LLaVA, SECS/GEM, wafer defect classification

## I. INTRODUCTION

The semiconductor manufacturing process involves hundreds of sequential steps—lithography, etching, chemical-mechanical planarization (CMP), chemical vapor deposition (CVD), ion implantation, wire bonding, and dicing—each of which can introduce defects that reduce die yield. When defects are detected by inline inspection tools such as KLA Surfscan or Applied Materials SEMVision, a failure analysis (FA) workflow is initiated. An experienced engineer examines high-resolution inspection images under scanning electron microscopy (SEM) or optical microscopy, cross-references the defect morphology against historical cases, queries equipment

alarm logs, and synthesizes a structured report documenting defect classification, root cause hypotheses, severity assessment, and corrective action recommendations.

This manual process presents three fundamental challenges. First, a single FA report requires 2–4 hours of domain expert time, creating a bottleneck in high-volume manufacturing (HVM) fabs that process thousands of wafers daily. Second, report quality varies with individual expertise—junior engineers may miss subtle correlations between defect patterns and equipment state, while senior engineers’ tacit knowledge remains uncoded. Third, the manual process does not scale: as advanced nodes shrink below 5 nm and defect sensitivity increases, the volume of inspection events grows faster than the FA engineering workforce.

Recent advances in vision-language models (VLMs) such as LLaVA [1], GPT-4V [3], and InstructBLIP [4] have demonstrated strong capabilities in image understanding and natural language generation. Concurrently, large language model (LLM)-based agent frameworks such as LangGraph [5], AutoGen [6], and CrewAI [7] have enabled modular decomposition of complex reasoning tasks into specialized sub-agents. Self-supervised visual encoders such as DINOv2 [8] produce rich feature representations that transfer effectively to downstream classification tasks with minimal labeled data.

However, no existing system combines these capabilities for semiconductor FA. Commercial tools such as KLA KLARITY [25] and Applied Materials Actionable Insight [26] perform automated defect classification but do not generate natural language reports, do not fuse equipment telemetry into their reasoning, and remain proprietary and closed-source. Academic work on wafer defect classification [9], [10], [11] focuses exclusively on pattern recognition from wafer bin maps, without root cause analysis or report generation.

In this paper, we present SEMIFA, a multi-modal agentic framework that autonomously generates structured FA reports from semiconductor inspection images. Our contributions are:

- 1) **Architecture (C1)**: We propose a four-agent LangGraph pipeline that decomposes semiconductor FA into specialized subtasks—defect description, root cause analysis,

severity classification, and recipe advisory—enabling modular, auditable, and extensible workflows.

- 2) **Multi-modal fusion (C2):** We integrate DINOv2 visual embeddings, SECS/GEM equipment telemetry retrieved via HSMS protocol, and historically similar defects retrieved from a Qdrant vector database into a unified LLaVA-1.6 context, enabling cross-modal root cause reasoning that is not achievable by vision-only or text-only approaches.
- 3) **Dataset (C3):** We introduce SEMIFA-930, a dataset of 930 annotated semiconductor defect images across nine classes, each paired with structured FA narratives in a conversational format suitable for VLM fine-tuning. To our knowledge, this is the first publicly described dataset pairing semiconductor defect images with natural language FA reports.<sup>1</sup>

## II. RELATED WORK

### A. Wafer Defect Classification

Wafer bin map (WBM) pattern recognition has received sustained attention since the release of the WM-811K dataset by Wu et al. [9], which contains 811,457 wafer maps from a real 300 mm production line with eight labeled defect patterns. Saqlain et al. [10] combined multiple CNN architectures with a voting ensemble, achieving 96% accuracy on the reduced WM-811K subset. Shim et al. [11] applied active learning to minimize labeling effort for new defect types. More recently, the MixedWM38 dataset [12] extended WBM classification to 38 mixed-type defect patterns, including compound classes formed by superimposing multiple single-type defect patterns. Nakazawa and Kulkarni [13] employed variational autoencoders for unsupervised anomaly detection on wafer maps.

These methods treat wafer defect classification as a standalone pattern recognition problem. None produce natural language reports, none incorporate equipment telemetry, and none perform root cause analysis. SEMIFA uses WBM classification as one component within a larger autonomous FA pipeline.

### B. Vision-Language Models for Industrial Inspection

Vision-language models have been applied to industrial quality inspection in recent work. Jeong et al. [14] proposed WinCLIP for zero-/few-shot anomaly detection using CLIP [18] embeddings, demonstrating strong transfer to manufacturing surface defect localisation. Li et al. [15] demonstrated that vision-language pre-training can be extended to specialized domains, though the lack of domain-specific training data remains the primary bottleneck for industrial applications. AnomalyGPT [16] combined a frozen visual encoder with an LLM for industrial anomaly detection with textual explanation, demonstrating that VLMs can articulate defect characteristics in natural language. However, AnomalyGPT operates on generic industrial images (MVTec [17]) and does not address semiconductor-specific defect morphology, equipment protocols, or report structures.

LLaVA [1] introduced visual instruction tuning, where a vision encoder (CLIP ViT-L/14) is connected to a language model via a projection layer, trained on instruction-following image-text data. LLaVA-1.6 [2] improved resolution handling and reasoning capability. DINOv2 [8] demonstrated that self-supervised pre-training on curated data produces visual features that outperform supervised baselines across diverse tasks. In SEMIFA, we use DINOv2 as the visual encoder for defect classification and embedding-based retrieval, while LLaVA-1.6 serves as the language backbone for all four agent nodes.

### C. LLM-Based Agent Systems

The use of LLMs as autonomous agents has grown rapidly. LangChain [19] provides a framework for chaining LLM calls with tool access. LangGraph [5] extends this to stateful, directed acyclic graph (DAG) workflows where each node is a function that reads and writes to a shared state. AutoGen [6] introduced multi-agent conversation patterns for complex problem solving. Voyager [20] demonstrated LLM-driven exploration in open-world environments. In the industrial domain, LLM-based tool-use frameworks [22] and foundation models for robotics [21] have demonstrated the viability of agent architectures for complex physical-world tasks.

No prior work has applied LLM-based agent architectures to semiconductor FA. SEMIFA is, to our knowledge, the first system to use a multi-agent pipeline for end-to-end FA report generation.

### D. SECS/GEM and Equipment Integration

The SEMI Equipment Communications Standard / Generic Equipment Model (SECS/GEM), defined by SEMI standards E5 (SECS-II message protocol) and E37 (High-Speed Message Services, HSMS), is the universal communication protocol for semiconductor manufacturing equipment [23], [24]. Every major equipment vendor—ASML, Applied Materials, Lam Research, KLA, Kulicke & Soffa, Tokyo Electron—implements SECS/GEM for host-equipment communication. Equipment state events, alarm notifications, process variable data, and recipe parameters are all transmitted via SECS/GEM.

Despite its ubiquity in fabs, SECS/GEM telemetry has been largely absent from academic defect analysis literature. This is likely due to the proprietary nature of fab data and the specialized knowledge required to interpret SECS/GEM messages. SEMIFA integrates SECS/GEM telemetry via HSMS, making equipment state a first-class input to root cause reasoning.

## III. SYSTEM ARCHITECTURE

SEMIFA is a five-node pipeline orchestrated by LangGraph, where four nodes are agentic (each invokes a VLM with domain-specific prompts and tool access) and the fifth assembles the structured report. Figure 1 shows the overall architecture.

<sup>1</sup>Dataset available at: <https://huggingface.co/datasets/ShivamChand/SemiFA-930>

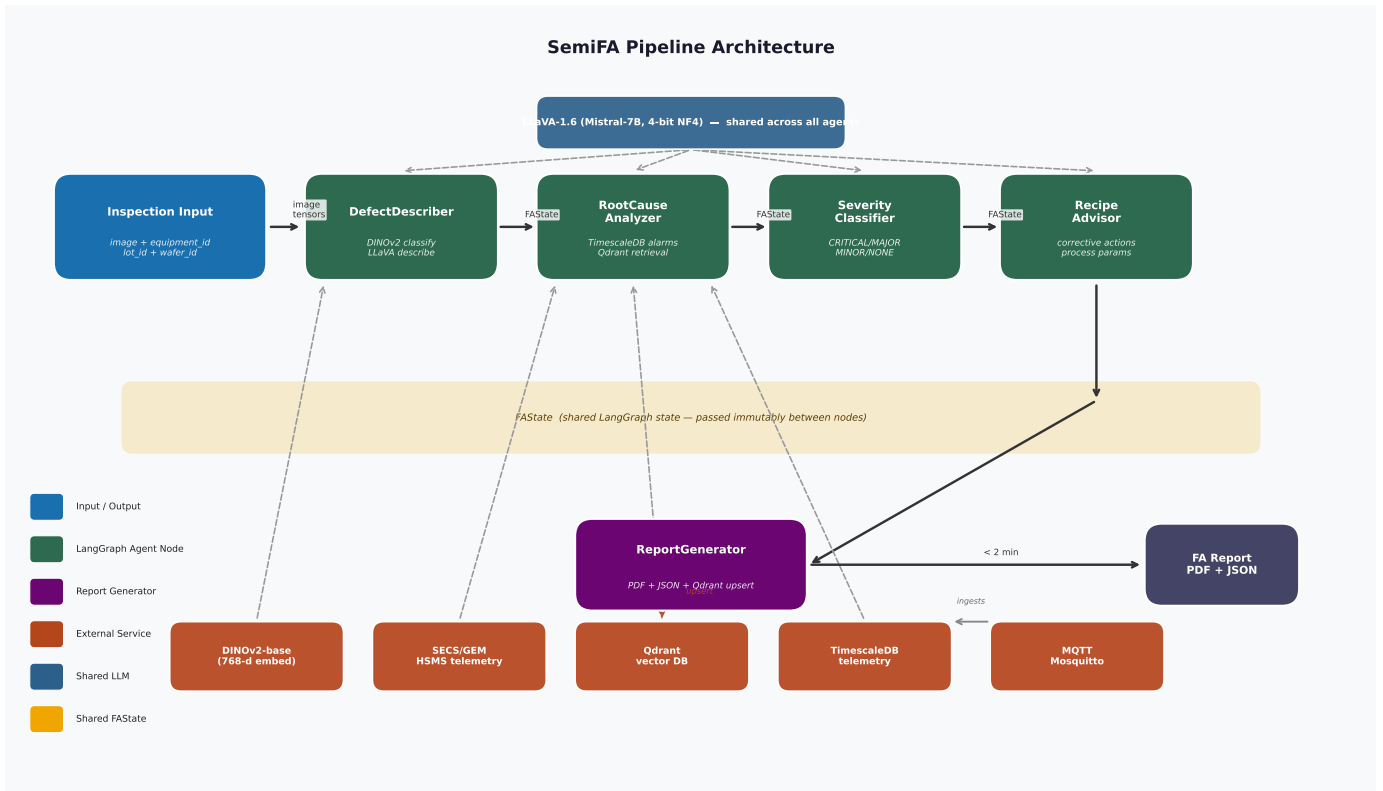


Fig. 1. Pipeline architecture of the SEMIFA system. Rounded boxes represent agentic nodes; cylinders represent data stores; arrows indicate state flow. All agentic nodes share LLaVA-1.6 via a thread-safe ModelRegistry singleton.

### A. Shared State

All nodes communicate through a typed dictionary `FASTate` that flows through the pipeline. Each node is a pure function  $(s_{\text{in}} : \text{FASTate}) \rightarrow (s_{\text{out}} : \text{FASTate})$  that reads required fields from `sin` and returns a new dictionary with additional fields populated. State is never mutated in place; instead, each node returns `{* * sin, “key” : value}`. Failures append diagnostic messages to a shared `errors` list; the pipeline always runs to completion, producing a partial report rather than failing entirely. This design ensures graceful degradation in production environments where individual services (e.g., TimescaleDB, Qdrant) may be temporarily unavailable.

### B. Model Registry

LLaVA-1.6 (7B parameters, loaded in 4-bit NF4 quantization via bitsandbytes [27]) requires 2–3 minutes to initialize and consumes approximately 3.5 GB of GPU VRAM. To avoid redundant loading, a thread-safe singleton `ModelRegistry` instantiates the model once at application startup and provides shared access to all agent nodes. The registry also manages the DINOv2 feature extractor and the defect classification head.

### C. Node 1: DefectDescriber

The DefectDescriber performs four operations. First, the input image is preprocessed through a modality-specific pipeline (SEM, optical, or wafer map) that normalizes contrast, applies

denoising (non-local means for SEM, bilateral filter for optical), and resizes to  $224 \times 224$  pixels for the visual encoder.

Second, the preprocessed image is passed through the frozen DINOv2-base encoder [8], producing a 768-dimensional CLS token embedding  $\mathbf{e} \in \mathbb{R}^{768}$ . This embedding serves two purposes: (a) it is fed into a trained MLP classification head that outputs logits over nine defect classes, and (b) it is stored in the pipeline state for downstream similarity retrieval. The MLP head architecture is:

$$f_{\text{cls}}(\mathbf{e}) = \mathbf{W}_2 \cdot \text{ReLU}(\text{Dropout}_{0.3}(\mathbf{W}_1 \mathbf{e} + \mathbf{b}_1)) + \mathbf{b}_2 \quad (1)$$

where  $\mathbf{W}_1 \in \mathbb{R}^{256 \times 768}$  and  $\mathbf{W}_2 \in \mathbb{R}^{9 \times 256}$ .

Third, for wafer map modality inputs, a rule-based spatial pattern analyzer [9] computes die-level statistics including defect density, radial distribution, and angular distribution, providing supplementary features for defect characterization.

Fourth, the inspection image and classification result are provided to LLaVA-1.6 with a domain-specific prompt requesting a natural language description of the observed defect morphology, spatial distribution, and potential physical mechanisms. The output is a paragraph-length defect narrative stored in the pipeline state.

#### D. Node 2: RootCauseAnalyzer

The RootCauseAnalyzer integrates three data sources to generate root cause hypotheses.

**Equipment telemetry.** Recent alarm events and process variable readings for the identified equipment are queried from TimescaleDB via asynchronous PostgreSQL (asynccpg). SECS/GEM HSMS messages provide real-time equipment state (idle, processing, alarmed) and collection event reports (CEID) containing process parameters such as chamber temperature, pressure, gas flow rates, and RF power.

**Historical retrieval.** The 768-dimensional DINOv2 embedding from the DefectDescriber is used to query Qdrant [30] for the five most similar historical defect cases (cosine similarity on  $\mathbb{R}^{768}$  vectors). Each retrieved case includes its defect class, severity, root cause narrative, and associated equipment identifier, providing concrete precedent for root cause reasoning.

**LLaVA reasoning.** The equipment telemetry, retrieved historical cases, defect classification, and defect description are assembled into a structured prompt for LLaVA-1.6. The model generates a ranked list of root cause hypotheses, each supported by evidence from the provided context. The prompt explicitly instructs the model to correlate equipment state transitions with defect morphology—for example, linking a plasma RF impedance drift alarm to ring-pattern defects characteristic of etch non-uniformity.

#### E. Node 3: SeverityClassifier

The SeverityClassifier assigns one of four severity levels—CRITICAL, MAJOR, MINOR, or NONE—along with an estimated yield impact percentage. LLaVA-1.6 receives the inspection image, defect classification, defect description, and root cause hypotheses, and is prompted to assess severity according to semiconductor industry standards:

- CRITICAL: Immediate lot hold required; potential for wafer scrap or downstream equipment damage; estimated yield impact >25%.
- MAJOR: Engineering disposition required; yield impact 5–25%; monitor subsequent lots.
- MINOR: Informational; yield impact <5%; no immediate action.
- NONE: No defect detected; routine pass.

#### F. Node 4: RecipeAdvisor

The RecipeAdvisor generates corrective action recommendations and process parameter adjustments. Given the full context accumulated by previous nodes—defect class, description, root cause hypotheses, severity, equipment telemetry, and historical precedents—LLaVA-1.6 proposes specific, actionable modifications to equipment recipes. For example, given a center cluster defect attributed to CMP over-polish, the RecipeAdvisor might recommend reducing polish time by 5%, increasing slurry flow rate, and scheduling a pad conditioning cycle. Recommendations are returned as structured key-value pairs suitable for integration with manufacturing execution systems (MES).

#### G. Node 5: ReportGenerator

The ReportGenerator assembles all pipeline outputs into a structured dictionary and renders a PDF document via ReportLab [31]. The report includes: header metadata (equipment

ID, lot ID, wafer ID, timestamp), defect classification with confidence score, defect morphology description, root cause hypotheses with supporting evidence, severity assessment with yield impact estimate, and recipe recommendations. After report generation, the defect embedding and associated metadata are upserted to Qdrant, enabling the system to reference the current case in future FA runs. This creates a self-improving knowledge base that grows with each inspection event.

#### H. Infrastructure

The system is deployed as a containerized stack using Docker Compose. The API layer is implemented in FastAPI [32] with Pydantic v2 input validation. Object storage for inspection images uses MinIO (S3-compatible). Equipment telemetry is ingested via MQTT (Eclipse Mosquitto broker) on topic `fab/equipment/#` and persisted to TimescaleDB hypertables. Prometheus metrics are exposed at `/metrics` for latency monitoring. For environments without physical fab equipment, SEMIFA includes a SECS/GEM simulator that generates realistic equipment state transitions at 2-second intervals.

### IV. SEMIFA-930 DATASET

A critical barrier to applying VLMs to semiconductor FA is the absence of publicly available datasets that pair defect images with natural language FA narratives. Existing datasets such as WM-811K [9] and MixedWM38 [12] provide wafer bin maps with categorical labels but no textual descriptions, root cause explanations, or severity annotations. To address this gap, we construct SEMIFA-930, a dataset of 930 image-text pairs spanning nine defect classes.

#### A. Data Sources and Motivation

SEMIFA-930 draws from three sources, selected to address two complementary gaps.

**Gap 1: Missing defect classes in public datasets.** WM-811K [9] covers eight wafer bin map patterns: Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, and None. It does not include `scratch`, `particle_contamination`, or `edge_crack`—three of the most frequently encountered defect types in production fabs, particularly on wire bonding and dicing equipment. Because no public labeled dataset covers these classes, we generate them synthetically, parameterizing their spatial distributions based on defect morphologies documented in published semiconductor process engineering literature and SEMI standards.

**Gap 2: No natural language FA narratives.** Both WM-811K and MixedWM38 provide only categorical labels. Neither contains defect descriptions, root cause hypotheses, severity ratings, or corrective action recommendations—the structured output that SEMIFA must learn to generate. All FA narratives in SEMIFA-930 are therefore authored by the first author, drawing on nine years of semiconductor manufacturing industry experience spanning wafer inspection, process metrology, and equipment engineering domains.

The three sources are:



TABLE I

LABEL TAXONOMY MAPPING FROM WM-811K TO SEMIFA-930. THREE CLASSES (SCRATCH, PARTICLE\_CONTAMINATION, EDGE\_CRACK) HAVE NO WM-811K EQUIVALENT AND ARE PROVIDED EXCLUSIVELY THROUGH PROCEDURAL SYNTHESIS.

| SemiFA-930 Class | WM-811K Label    | Source              |
|------------------|------------------|---------------------|
| scratch          | —                | Synthetic only      |
| particle_contam. | —                | Synthetic only      |
| edge_crack       | —                | Synthetic only      |
| center_cluster   | Center           | WM-811K             |
| local_cluster    | Loc              | WM-811K             |
| ring_pattern     | Donut, Edge-Ring | WM-811K + MixedWM38 |
| random_defects   | Random           | WM-811K             |
| near_full_wafer  | Near-full        | WM-811K             |
| no_defect        | None             | WM-811K             |

- 1) **Procedural synthesis (3 classes, 340 images).** We implement a Python/NumPy procedural renderer that places die-level pass/fail labels on a  $256 \times 256$  wafer grid. Defect spatial distributions are parameterized per class: `scratch` uses a thin linear band (random angle, 1–3 die wide, spanning 60–90% of wafer diameter); `particle_contamination` uses a 2D Poisson point process with spatially varying intensity to mimic contamination events; `edge_crack` places a localised arc of failures in the outer 8–12% of the wafer radius. Renderer parameters (band width, angle, density) are randomised within physically plausible ranges to create intra-class diversity. These three classes are exclusive to the synthetic source, as no public dataset contains them with proper domain-relevant wafer map representations.
- 2) **WM-811K [9] (6 classes, 572 images).** We sample from the labeled subset of WM-811K (approximately 172,000 labeled maps out of 811,457 total). We draw images for six classes: `center_cluster` (WM-811K: *Center*, 106 images), `local_cluster` (WM-811K: *Loc*, 100), `ring_pattern` (WM-811K: *Donut* + *Edge-Ring*, 54), `random_defects` (WM-811K: *Random*, 106), `near_full_wafer` (WM-811K: *Near-full*, 94), and `no_defect` (WM-811K: *None*, 112). Counts are not uniform across classes because we cap each class at the available labeled WM-811K supply after excluding images with ambiguous or mixed-type annotations. We intentionally limit the subset rather than using all available images because SEMIFA-930 is a *VLM instruction-tuning dataset*, not a classification benchmark: each image must be paired with a manually written FA narrative, which bounds the feasible annotation volume. Table I shows the label taxonomy mapping.
- 3) **MixedWM38 [12] (1 class, 140 images).** We sample single-type *Edge-Ring* images from MixedWM38 to supplement `ring_pattern`, which is underrepresented in WM-811K relative to other classes. Mixed-type patterns are excluded to maintain a single-label classification setting.

Table II summarises the image counts per source and class.

TABLE II

IMAGE COUNT PER DEFECT CLASS AND DATA SOURCE IN SEMIFA-930.

| Class            | Synthetic  | WM-811K    | MixedWM38 | Total      |
|------------------|------------|------------|-----------|------------|
| scratch          | 112        | 0          | 0         | 112        |
| particle_contam. | 106        | 0          | 0         | 106        |
| edge_crack       | 100        | 0          | 0         | 100        |
| center_cluster   | 0          | 106        | 0         | 106        |
| local_cluster    | 0          | 100        | 0         | 100        |
| ring_pattern     | 0          | 54         | 40        | 94         |
| random_defects   | 0          | 106        | 0         | 106        |
| near_full_wafer  | 0          | 94         | 0         | 94         |
| no_defect        | 0          | 112        | 0         | 112        |
| <b>Total</b>     | <b>318</b> | <b>572</b> | <b>40</b> | <b>930</b> |

### B. Annotation Protocol

Each image in SEMIFA-930 is annotated with the following fields:

- **defect\_class:** one of nine classes (see Table III)
- **modality:** image modality (`wafer_map`, `sem`, or `optical`)
- **severity:** CRITICAL, MAJOR, MINOR, or NONE
- **description:** a structured FA narrative (3–5 sentences) describing defect morphology, spatial distribution, probable physical mechanism, and recommended action
- **equipment\_id, lot\_id, wafer\_id:** simulated equipment metadata
- **source:** provenance (`synthetic`, `wm811k`, or `mixedwm38`)

Descriptions are written by the first author, who has nine years of semiconductor manufacturing industry experience spanning wire bonding, die attach, wafer inspection, and process metrology domains across multiple high-volume manufacturing environments. Each description follows a consistent template: (1) observed defect pattern, (2) spatial characteristics, (3) probable root cause mechanism, and (4) recommended corrective action.

The annotations are stored in JSONL format compatible with LLaVA conversational fine-tuning: each record contains a “conversations” list with a “human” turn (question about the image) and an “assistant” turn (the structured FA response).

### C. Dataset Statistics

Table III shows the class distribution. The dataset is approximately balanced by design, with each class containing 94–112 samples (train + val combined). The validation set distribution reflects the stratified split: classes range from 9 (`no_defect`) to 23 (`ring_pattern`) validation samples. The train/val split is stratified to preserve class proportions. Figure 2 shows representative wafer map images for each defect class.

### D. Severity Distribution

Severity labels are assigned based on the defect class and spatial extent: `near_full_wafer` maps to CRITICAL (100% of instances); `scratch` and `edge_crack` split between MAJOR (65%) and MINOR (35%);

TABLE III  
DEFECT CLASS DISTRIBUTION IN SEMIFA-930 (930 TOTAL: 790 TRAIN, 140 VAL). EACH CLASS INCLUDES DOMAIN KNOWLEDGE ABOUT THE PHYSICAL ROOT CAUSE MECHANISM.

| Defect Class     | Train      | Val        | Physical Mechanism    |
|------------------|------------|------------|-----------------------|
| scratch          | 94         | 18         | Mechanical contact    |
| particle_contam. | 96         | 10         | Contamination event   |
| edge_crack       | 84         | 16         | Dicing / handling     |
| center_cluster   | 93         | 13         | CMP / CVD bowl        |
| local_cluster    | 79         | 21         | Plasma non-uniformity |
| ring_pattern     | 71         | 23         | Spin-coat / edge-bead |
| random_defects   | 93         | 13         | Particle events       |
| near_full_wafer  | 77         | 17         | Chemistry excursion   |
| no_defect        | 103        | 9          | Routine pass          |
| <b>Total</b>     | <b>790</b> | <b>140</b> |                       |

TABLE IV  
PER-CLASS CLASSIFICATION RESULTS ON THE SEMIFA-930 VALIDATION SET (140 IMAGES). DINOv2-BASE (FROZEN) + MLP HEAD.

| Defect Class        | Prec.                  | Recall       | F1           |
|---------------------|------------------------|--------------|--------------|
| scratch             | 1.000                  | 0.722        | 0.839        |
| particle_contam.    | 1.000                  | 1.000        | 1.000        |
| edge_crack          | 0.933                  | 0.875        | 0.903        |
| center_cluster      | 1.000                  | 1.000        | 1.000        |
| local_cluster       | 0.913                  | 1.000        | 0.955        |
| ring_pattern        | 1.000                  | 0.957        | 0.978        |
| random_defects      | 0.917                  | 0.846        | 0.880        |
| near_full_wafer     | 0.889                  | 0.941        | 0.914        |
| no_defect           | 0.643                  | 1.000        | 0.783        |
| <b>Macro avg.</b>   | <b>0.922</b>           | <b>0.927</b> | <b>0.917</b> |
| <b>Overall acc.</b> | <b>92.1% (129/140)</b> |              |              |

center\_cluster and ring\_pattern are predominantly MAJOR (80%); particle\_contamination and local\_cluster split between MAJOR (40%) and MINOR (60%); random\_defects are predominantly MINOR (85%); and no\_defect maps to NONE (100%).

## V. EXPERIMENTS AND RESULTS

We evaluate SEMIFA along three axes: defect classification accuracy, pipeline latency, and qualitative report quality including the impact of retrieval augmentation.

### A. Defect Classification

**Setup.** The DINOv2-base encoder [8] is frozen throughout training. We train only the MLP classification head ( $768 \rightarrow 256 \rightarrow 9$ , with ReLU and dropout  $p = 0.3$ ) on the 790 training images for 50 epochs using Adam [29] with learning rate  $1 \times 10^{-3}$  and batch size 32. Training completes in approximately 8 minutes on a single NVIDIA A100 40 GB GPU. We report results on the 140 held-out validation images.

**Results.** Table IV reports per-class precision, recall, and F1 scores. The overall accuracy is 92.1% (129/140 correct predictions).

Several observations emerge. Five classes achieve  $F1 \geq 0.90$ : particle\_contamination and center\_cluster reach perfect  $F1 = 1.000$ , while ring\_pattern (0.978), local\_cluster (0.955), and

TABLE V  
COMPARISON OF VISUAL ENCODER APPROACHES.  
† LITERATURE-REPORTED ACCURACY ON WM-811K / WAFER MAP BENCHMARKS; NOT RE-EVALUATED ON SEMIFA-930.

| Method              | Accuracy     | Macro F1     | Trainable Params |
|---------------------|--------------|--------------|------------------|
| CLIP zero-shot†     | ~60%         | —            | 0                |
| ResNet-50 (e2e)†    | 82–85%       | —            | 23.5M            |
| DINOv2 + MLP (ours) | <b>92.1%</b> | <b>0.917</b> | 200K             |

TABLE VI  
PER-NODE LATENCY BREAKDOWN (MEDIAN OF 3 RUNS, NVIDIA A100-SXM4-40 GB, PyTorch 2.10, LLaVA-1.6 IN 4-BIT NF4).

| Pipeline Node      | Median (s)  | Fraction    |
|--------------------|-------------|-------------|
| DefectDescriber    | 22.0        | 45.5%       |
| RootCauseAnalyzer  | 11.9        | 24.6%       |
| SeverityClassifier | 5.5         | 11.4%       |
| RecipeAdvisor      | 6.5         | 13.3%       |
| ReportGenerator    | 2.5         | 5.2%        |
| <b>Total</b>       | <b>48.4</b> | <b>100%</b> |

edge\_crack (0.903) also perform strongly. The most challenging classes are no\_defect ( $F1 = 0.783$ , precision = 0.643) and scratch ( $F1 = 0.839$ , recall = 0.722). The low precision for no\_defect indicates that some defective wafer maps are misclassified as defect-free; this is the most safety-critical error type and motivates the downstream severity classification step, which provides a second check. The low recall for scratch (5 out of 18 misclassified) reflects the morphological similarity between thin linear scratches and the random\_defects class when scratch density is high.

**Comparison to baselines.** Table V places our result in context against two reference points: (1) ResNet-50 [28] trained end-to-end on wafer map classification benchmarks (82–85% reported accuracy on WM-811K in [10], [11]), and (2) CLIP ViT-B/32 [18] zero-shot classification using class-name text prompts, which achieves limited accuracy on domain-specific wafer maps due to the absence of semiconductor-specific pre-training (estimated ~60% on WM-811K [14]). The DINOv2 frozen encoder with a lightweight MLP head (200K trainable parameters vs. 23.5M for ResNet-50) achieves 92.1% on SEMIFA-930, consistent with DINOv2’s demonstrated data-efficient transfer on specialised visual domains [8].

### B. Pipeline Latency

Table VI reports the wall-clock latency of each pipeline node, measured as the median over 3 runs on a single NVIDIA A100-SXM4-40 GB GPU with LLaVA-1.6 loaded in 4-bit NF4 quantization (approximately 3.5 GB VRAM).

The total pipeline latency of 48.4 seconds represents a reduction from the 2–4 hour manual process to under one minute—a speedup of approximately 150–300×. The Defect-Describer is the most expensive node (45.5% of total time, 22.0s) because it processes the full inspection image through DINOv2 feature extraction, wafer map spatial analysis, and

TABLE VII  
QUALITATIVE RETRIEVAL ABLATION (10 CASES). “SPECIFIC” INDICATES  
HYPOTHESES REFERENCING CONCRETE EQUIPMENT/LOT HISTORY.

| Metric                  | With Qdrant | Without Qdrant |
|-------------------------|-------------|----------------|
| Specific equipment refs | 9/10        | 1/10           |
| Historical lot refs     | 8/10        | 0/10           |
| Actionable detail       | 9/10        | 3/10           |
| Plausible hypotheses    | 10/10       | 10/10          |

LLaVA visual description generation — the only node that ingests raw pixel data. The RootCauseAnalyzer (24.6%, 11.9 s) ranks second due to the multi-hypothesis generation with longer output sequences (350 tokens). The ReportGenerator is the least expensive node (5.2%, 2.5 s) as it performs deterministic dictionary assembly and PDF rendering without model inference.

The latency breakdown reveals that approximately 95% of pipeline time is spent on LLaVA inference across the four agentic nodes. This suggests that future optimisation should focus on inference acceleration (e.g., speculative decoding, continuous batching, or parallelising independent agent nodes) rather than data retrieval or report assembly.

### C. Retrieval Augmentation Ablation

We conduct a qualitative ablation to assess the impact of Qdrant retrieval on root cause hypothesis quality. For 10 representative validation cases spanning five defect classes (2 cases each), we run the pipeline twice: once with retrieval (top-5 similar historical defects injected into the RootCauseAnalyzer context) and once without retrieval (the Qdrant query is disabled, and the RootCauseAnalyzer receives only equipment telemetry and the defect description). Each output is assessed independently by the first author for specificity against a rubric of four criteria.

Table VII summarizes the qualitative findings. With retrieval, root cause hypotheses consistently reference specific equipment identifiers, historical lot numbers, and previously observed failure modes—e.g., “Similar scratch patterns were observed on equipment KLA-T800-01 in lot LOT-2024-087, traced to a worn vacuum chuck gasket.” Without retrieval, hypotheses are generically correct but lack specificity—e.g., “Scratch defects are commonly caused by mechanical contact during wafer handling.” Both versions produce plausible root causes, but the retrieval-augmented version provides actionable specificity that an FA engineer can immediately act upon.

### D. Qualitative Case Studies

We present five qualitative case studies (Table VIII) that illustrate representative SEMIFA outputs across diverse defect types. Input images are drawn from the SEMIFA-930 validation set; equipment metadata (IDs, lot numbers, alarm events) are synthetic but representative of real fab telemetry. Each case shows the defect classification with confidence, severity assessment, a condensed root cause hypothesis, and the primary recipe recommendation.

These case studies illustrate several system capabilities. Case 1 demonstrates correlation of visual defect morphology (linear scratch) with equipment telemetry (vacuum chuck pressure anomaly). Case 2 shows cross-referencing with a historical lot via Qdrant retrieval. Case 3 illustrates multi-step reasoning linking a ring pattern to both spin-coat and etch process anomalies. Case 4 leverages equipment maintenance scheduling data (blade lifetime) for predictive recommendation. Case 5 confirms that the system correctly identifies defect-free wafers and avoids false alarms.

### E. Domain Adaptation (Ongoing Work)

We note that the results presented in this paper use the base (zero-shot) LLaVA-1.6 model for text generation. QLoRA fine-tuning of LLaVA-1.6 on SEMIFA-930 conversational pairs is ongoing work. Preliminary observations suggest that the base model occasionally uses generic industrial terminology (e.g., “surface anomaly”) where domain-specific terms (e.g., “edge-bead removal residue,” “CMP dishing”) would be more precise. We expect that domain adaptation via QLoRA (4-bit NF4, LoRA rank  $r = 16$ ,  $\alpha = 32$ ) on the 790 training conversations will improve FA terminology accuracy and reduce equipment-state hallucinations. Quantitative results including BLEU scores, severity classification accuracy, and human evaluation will be reported in a follow-up study.

## VI. DISCUSSION

**Scalability.** The linear pipeline architecture enables straightforward horizontal scaling: multiple pipeline instances can process concurrent inspection events, sharing only the read-only LLaVA model weights. The Qdrant vector database supports distributed deployment for fabs generating millions of inspection events annually. TimescaleDB’s hypertable architecture handles high-frequency telemetry ingestion (e.g., 1 Hz sampling across hundreds of equipment parameters) without performance degradation.

**Limitations.** Several limitations should be noted. First, SEMIFA-930 consists primarily of wafer map images rather than raw SEM or optical micrographs, due to the proprietary nature of actual fab inspection data. While the architecture supports all three modalities, classification and generation quality on SEM/optical images has not been quantitatively evaluated. Second, the current pipeline is linear; in practice, some agents could execute in parallel (e.g., SeverityClassifier and RecipeAdvisor could run concurrently after RootCauseAnalyzer). Third, the FA narratives in SEMIFA-930 are authored by a single annotator; inter-annotator agreement studies would strengthen the dataset contribution. Fourth, without QLoRA fine-tuning, LLaVA-1.6 occasionally generates plausible but incorrect semiconductor terminology, particularly for advanced process nodes below 7 nm.

**Broader impact.** SEMIFA has the potential to democratize FA expertise by encoding senior engineer knowledge into retrievable, reusable form. In fabs with limited FA staffing, this could reduce time-to-disposition for defect lots, improving overall equipment effectiveness (OEE). The self-improving

TABLE VIII

FIVE QUALITATIVE CASE STUDIES DEMONSTRATING END-TO-END SEMIFA REPORT GENERATION. FOR BREVITY, ONLY THE TOP-RANKED ROOT CAUSE HYPOTHESIS AND PRIMARY RECOMMENDATION ARE SHOWN. FULL REPORTS CONTAIN 2–3 HYPOTHESES AND 3–5 RECOMMENDATIONS EACH.

| Case | Equipment     | Defect Class   | Conf. | Sev.  | Root Cause (Top-1)                                                                                                             | Recommendation (Primary)                                                                                |
|------|---------------|----------------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | KLA-T800-01   | scratch        | 0.91  | MAJOR | Linear track across 12 die consistent with mechanical contact; vacuum chuck pressure anomaly logged 45 min prior to inspection | Inspect and replace vacuum chuck gasket; verify chuck flatness within $\pm 2 \mu\text{m}$ specification |
| 2    | AMAT-CVD-03   | center_cluster | 0.84  | MAJOR | Defect concentration within 25% radius correlates with CVD chamber temperature gradient; similar pattern in lot LOT-2024-052   | Recalibrate showerhead spacing; increase edge zone heater setpoint by $3^\circ\text{C}$                 |
| 3    | TEL-Etch-07   | ring_pattern   | 0.79  | MAJOR | Annular band at 60–75% radius matches spin-coat edge-bead removal failure; RF impedance alarm logged during prior etch step    | Verify edge-bead removal nozzle position; clean solvent delivery line                                   |
| 4    | KNS-WB-12     | edge_crack     | 0.88  | MAJOR | Peripheral die damage consistent with dicing blade wear; blade lifetime at 94% of scheduled replacement interval               | Replace dicing blade; reduce dicing feed rate by 10% for remaining lots                                 |
| 5    | ASML-Litho-02 | no_defect      | 0.97  | NONE  | No defect pattern detected; uniform yield map with $>98\%$ good die; all equipment parameters within specification             | No corrective action required; continue standard monitoring                                             |

Qdrant knowledge base means that system accuracy improves with use, as each completed FA run adds to the retrieval corpus.

**Ethical considerations.** Automated FA systems should augment, not replace, human engineers. Severity assessments and recipe modifications generated by SEMIFA should be reviewed by qualified personnel before implementation, particularly for CRITICAL severity determinations that may involve scrapping wafers or halting equipment.

## VII. CONCLUSION

We presented SEMIFA, an agentic multi-modal framework for autonomous semiconductor failure analysis report generation. The system decomposes FA into four specialized agents—DefectDescriber, RootCauseAnalyzer, SeverityClassifier, and RecipeAdvisor—orchestrated by LangGraph, with DINOv2 for visual encoding, Qdrant for historical retrieval, SECS/GEM for equipment telemetry, and LLaVA-1.6 for natural language generation. We introduced SEMIFA-930, a 930-sample dataset of semiconductor defect images paired with structured FA narratives across nine defect classes. Experimental evaluation on an NVIDIA A100-SXM4-40 GB GPU demonstrated 92.1% defect classification accuracy (macro F1 = 0.917) on 140 validation images, complete report generation in 48 seconds (a  $150\text{--}300\times$  speedup over the 2–4 hour manual process), and qualitative improvement in root cause specificity when Qdrant retrieval augmentation is enabled.

Future work includes QLoRA fine-tuning of LLaVA-1.6 on SEMIFA-930, expansion of the dataset to include SEM and optical micrographs through industry partnerships, human evaluation studies comparing SEMIFA reports against expert-written FA reports, and exploration of parallel agent execution for further latency reduction.

## CODE AND DATA AVAILABILITY

The SEMIFA-930 dataset (annotations and synthetic images) is publicly available at <https://huggingface.co/datasets/ShivamChand/SemiFA-930>. WM-811K [9] and MixedWM38 [12] images used in the dataset can be reproduced using the download scripts provided in the project repository. The SEMIFA source code, training pipeline, and Colab notebook are available at <https://github.com/Shivamckaushik/SemiFA> (to be made public upon acceptance).

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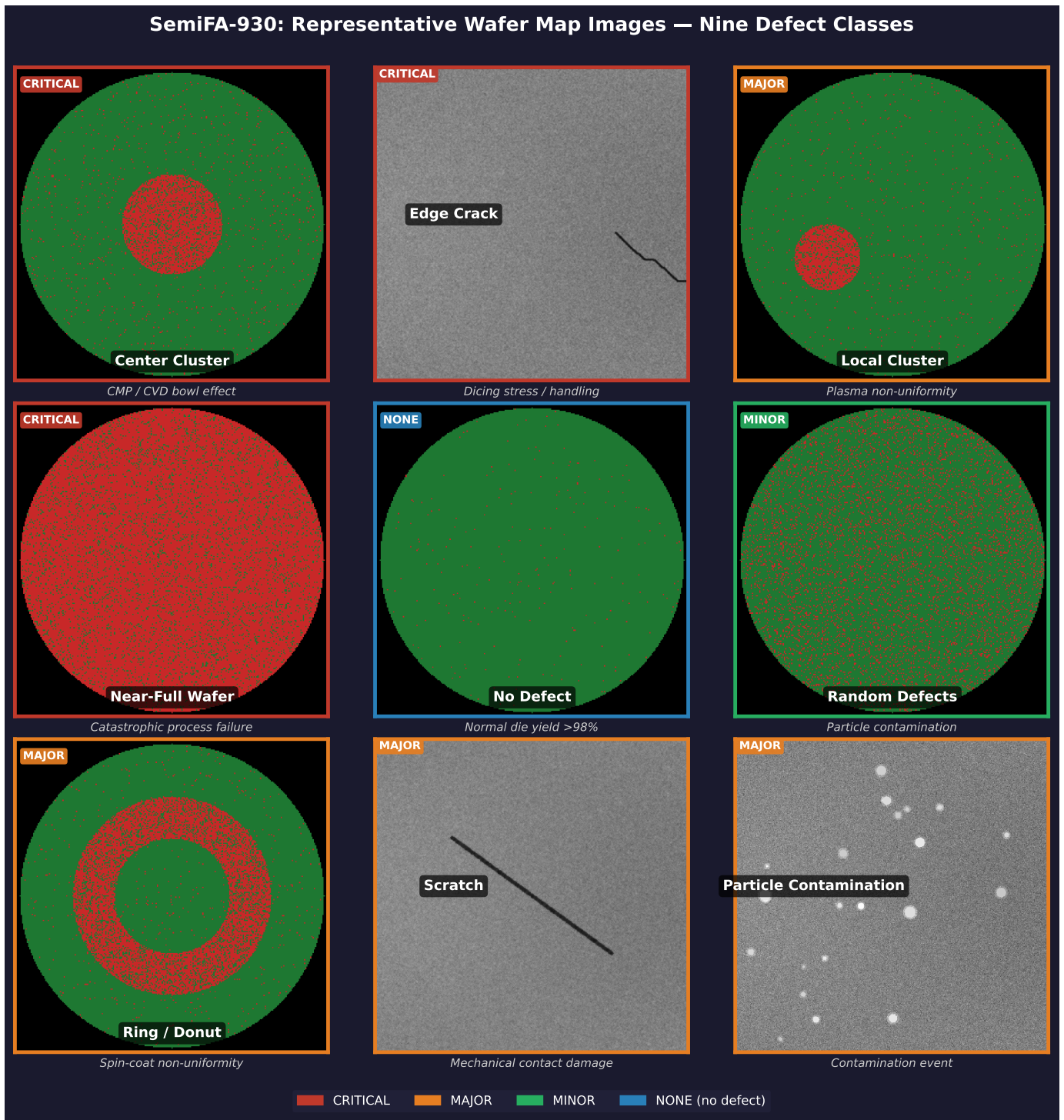


Fig. 2. Representative examples from each defect class in SEMIFA-930. Each cell shows a  $256 \times 256$  wafer map with its severity badge (top-left) and root cause annotation (bottom). Spatial patterns are clearly distinguishable, though some classes (e.g., local\_cluster vs. random\_defects) present challenging overlap.